AI-based Employee Wellness Management Platform

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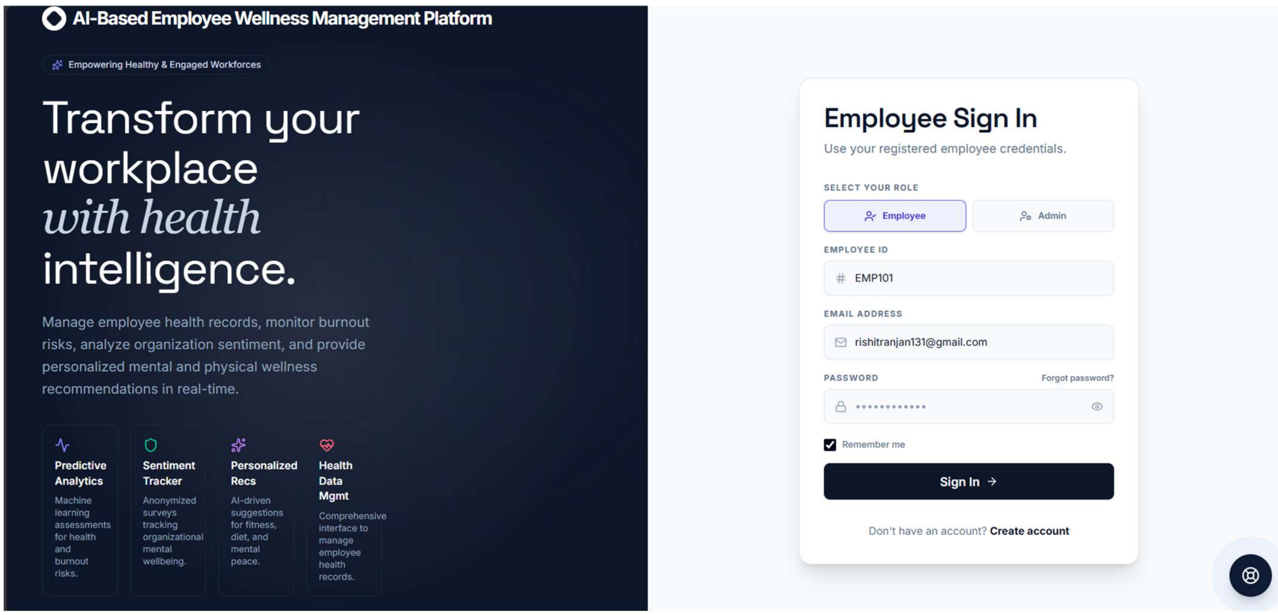
Transform workplace well-being with our Secure, Intelligent, and Role-Based Employee Wellness Platform. Powered by AI/ML, it delivers personalized recommendations, predictive risk analysis, and sentiment analytics to foster a thriving, resilient workforce.



Project Documentation

A full-stack web application for secure authentication, employee health records, wellness risk prediction, personalized recommendations, sentiment analysis, performance analytics, and AI wellness assistance.

Project Statement

Workplace wellness is often treated reactively, not proactively. Organizations frequently struggle to move beyond generic, one-size-fits-all programs that fail to engage employees or prevent burnout before it happens.

Existing wellness solutions are often fragmented, offering static dashboards and generic advice that lack personalization. This leaves administrators without actionable, predictive insights and employees without the tailored support they need to truly thrive.

This project addresses the critical need for a proactive and intelligent wellness ecosystem. It provides a unified, secure platform that leverages data analytics and machine learning to transform employee health from a reactive concern into a strategic, data-driven priority.

The system delivers personalized dashboards for employees with AI-driven recommendations and goal tracking, while providing administrators with predictive risk analysis, department-level sentiment analytics, and comprehensive KPI monitoring. With role-based access and an integrated AI wellness assistant, the platform aims to foster a healthier, more resilient, and productive workforce.

1. Project Overview and Outcomes

The application combines authentication, wellness data management, analytics, and intelligent assistance in a single web platform.

- **Secure and Role-Based Access**: Deliver secure employee and administrator registration and login, leading to distinct, role-based dashboards that tailor the user experience and protect sensitive data.
- **Centralized & Holistic Data Management**: Establish a single, unified platform for storing and managing all wellness data, including health records, daily habits, mental health logs, and sentiment pulses, eliminating data fragmentation.
- **Proactive Risk Identification & Intervention**: Implement an ML-powered risk prediction module to proactively identify at-risk employees, allowing administrators to move from a reactive to a preventative wellness strategy.
- **Increased Employee Engagement through Personalization**: Empower employees with a personalized dashboard, AI-driven wellness recommendations, and structured goal-setting tools to increase engagement and ownership of their health.
- **Actionable, Data-Driven Insights for Administrators**: Provide administrators with comprehensive analytics dashboards for monitoring department sentiment, program effectiveness, and key performance indicators (KPIs) like absenteeism and overall health risk.
- **On-Demand Intelligent Support**: Offer employees 24/7 access to an AI wellness chatbot for instant, contextual guidance on exercise, sleep, diet, and stress management, reducing reliance on unverified information.
- **Streamlined Administrative and Safety Processes**: Automate and simplify workflows for health check-up scheduling, insurance claims, and expense reimbursement, and implement an SOS feature for rapid emergency response.
- **A Scalable and Extensible Foundation**: Develop a robust, well-documented REST API that supports all platform features and is designed to be easily extended with future modules or integrated with other systems.

Primary Users

Employees manage their wellness profile, health vitals, habits, feedback, and recommendations. Administrators manage records, risk views, recommendations, sentiment analytics, and organizational KPIs.

2. Modules to be Implemented

Module 1: Authentication and Profile Management

Registration, login, logout, password recovery and role-based access

Module 2: Employee Wellness Profile

Self-reported BMI, blood pressure, exercise, sleep, stress level, mood, water, steps, and streak tracking.

Module 3: Health Records and Risk Prediction

Health record creation and updates, risk scoring, severity filtering, contributing factors, and prescribed actions.

Module 5: Sentiment and Performance Analytics

Department sentiment distribution, average stress score, participation rate, absenteeism, overall health risk, and program effectiveness.

Module 6: AI Wellness Chatbot

AI wellness assistant for exercise, sleep, diet and other wellness assistant

Module 7: Extension Modules

Notification center, insurance management, check-ups, SOS alerts, and health expense tracking, AI Wellness Coach, Diet Plans, etc.

3. Methodology

The platform is developed incrementally using a modular full-stack approach. Each layer is implemented and tested before connecting it to the next layer.

Step 1: Requirements and Module Planning

Identify employee/admin workflows, wellness data fields, authentication requirements, analytics needs, and future extension modules.

Step 2: Frontend Development

Build React + Vite screens, forms, dashboards, navigation, loading states, validation, and responsive components.

Step 3: Backend API Development

Create Flask endpoints for authentication, password reset, profiles, avatar upload, health records, and wellness services.

Step 4: Database and Security Integration

Connect MongoDB collections and apply bcrypt password hashing, JWT cookie authentication, role checks, validation, and duplicate prevention.

Step 5: Intelligent Processing

Load risk and recommendation models, calculate risk indicators, process sentiment using NLTK, and provide chatbot responses.

Step 6: Testing and Deployment Preparation

Test user journeys, API responses, role permissions, data persistence, error handling, and prepare the application for deployment.

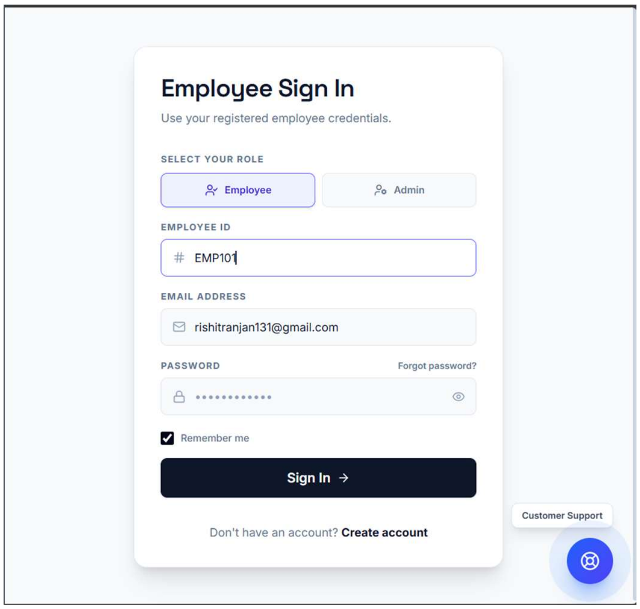
4. Week-wise Module Implementation Plan

Milestone 1: Weeks 1–2 — Login and Authentication

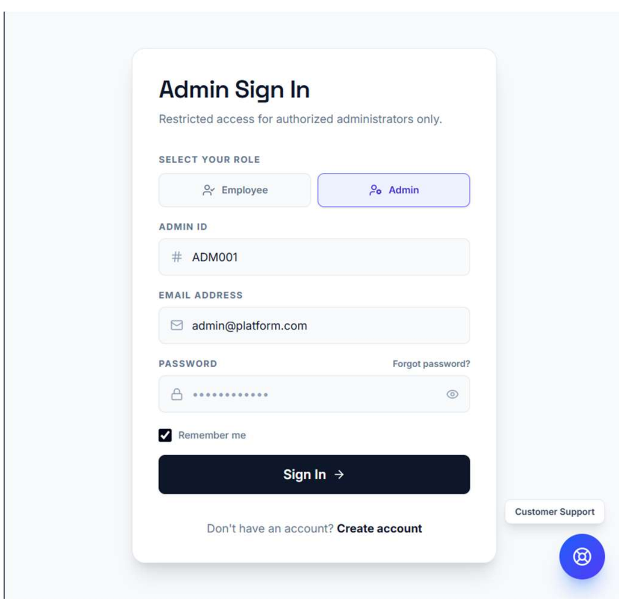
Goals: Implement registration, login, logout, password recovery, role handling, profile loading, and avatar support.

Tools: React, Flask, JWT, bcrypt, MongoDB

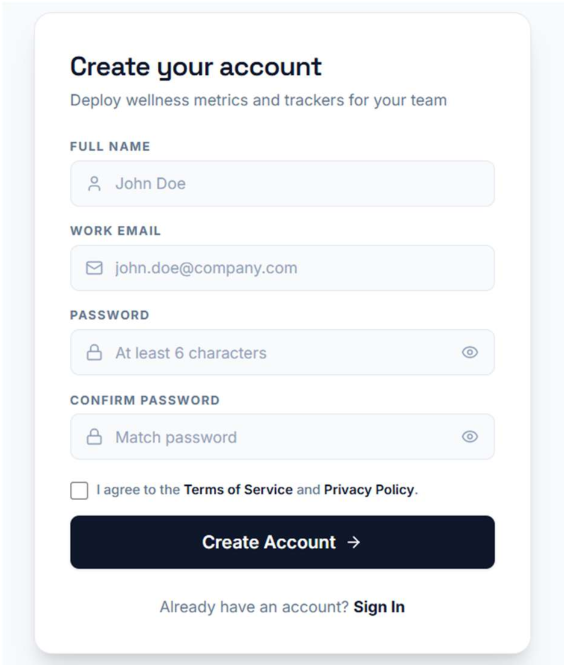
Employee Sign In



Admin Sign In



Account Creation

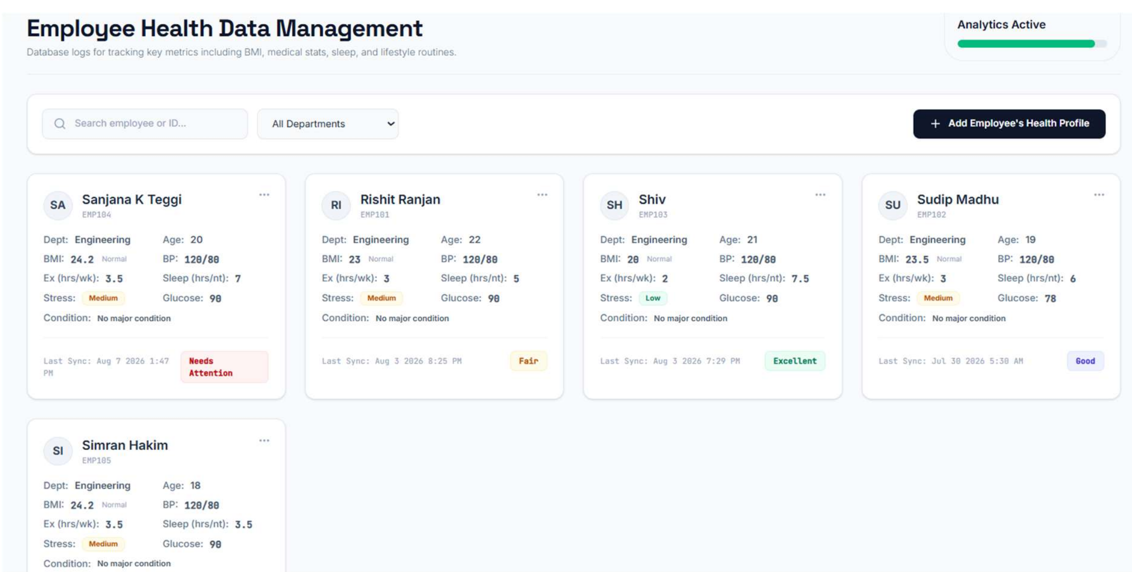


Milestone 2: Weeks 3–4 — Wellness Data

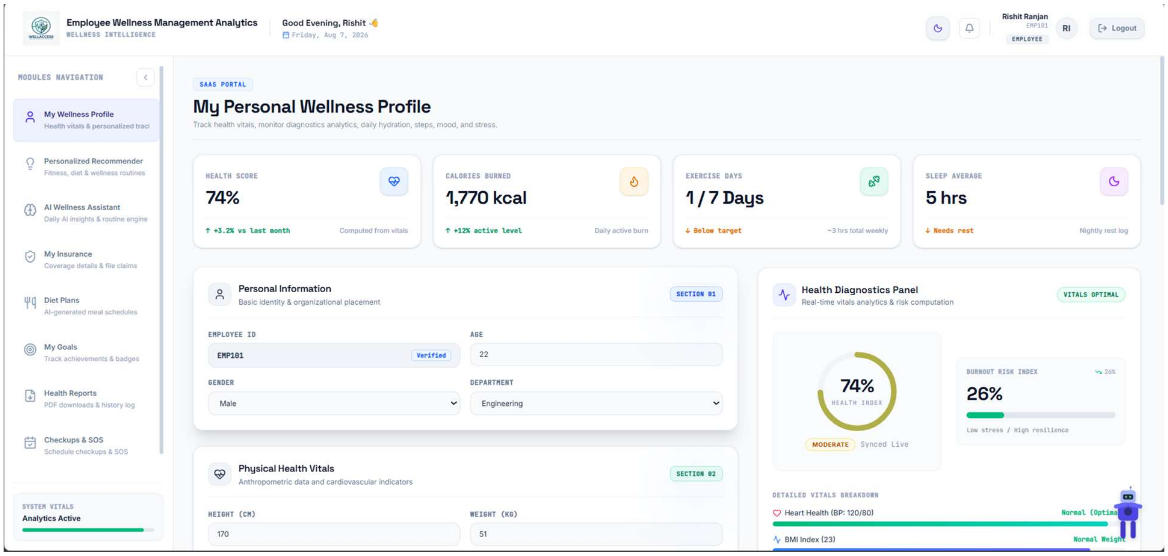
Goals: Implemented health records, employee profile vitals, habits, mood, feedback, and admin record management.

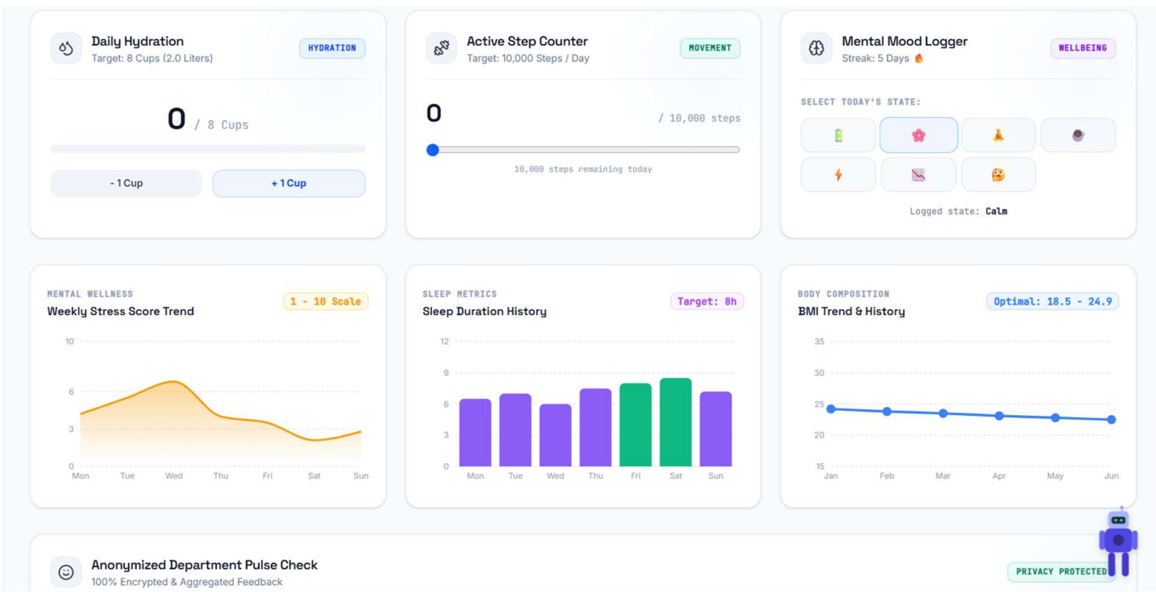
Tools: React forms, Flask REST API, PyMongo

Health Record Collection



Employee Wellness Profile



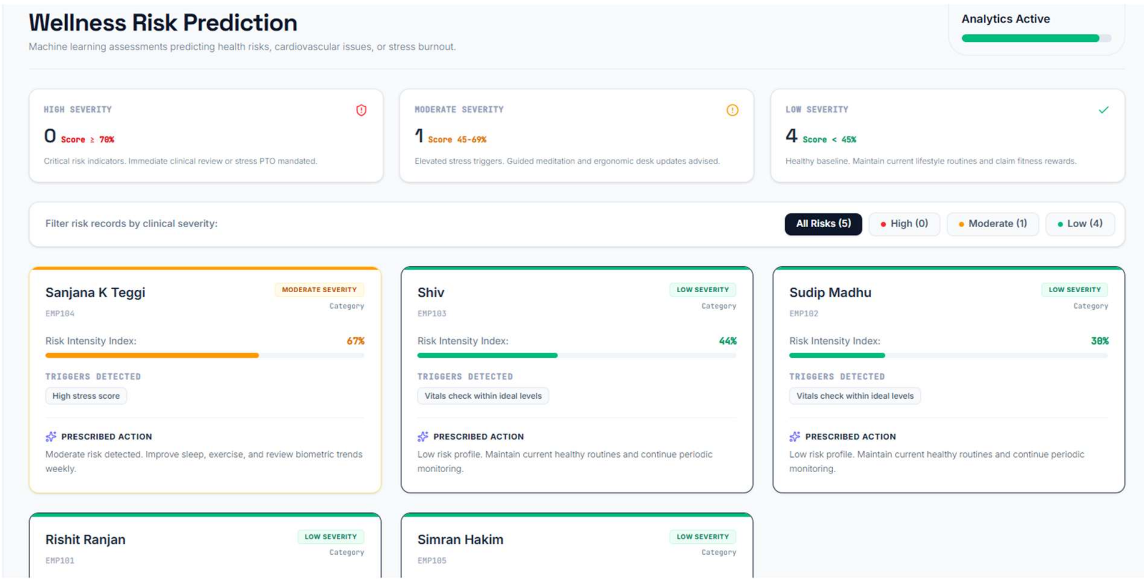


Milestone 3: Weeks 5–6 — Intelligence and Analytics

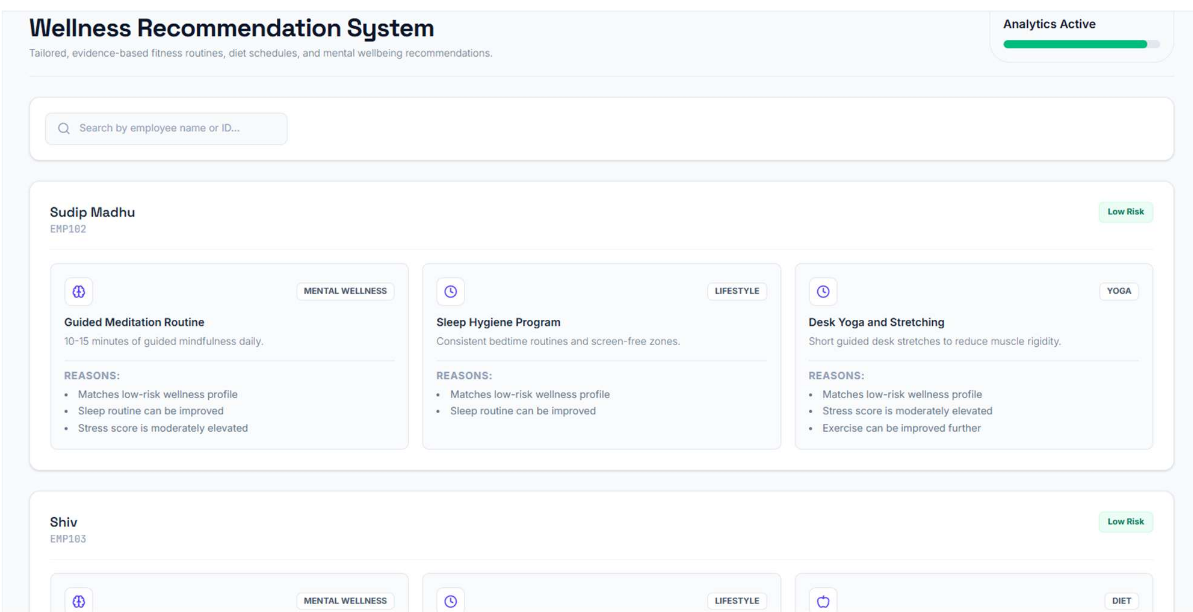
Goals: Add risk scoring, model loading, recommendation modules, sentiment analysis, KPI calculations, and chatbot interaction.

Tools: joblib, cloudpickle, NLTK, Python analytics

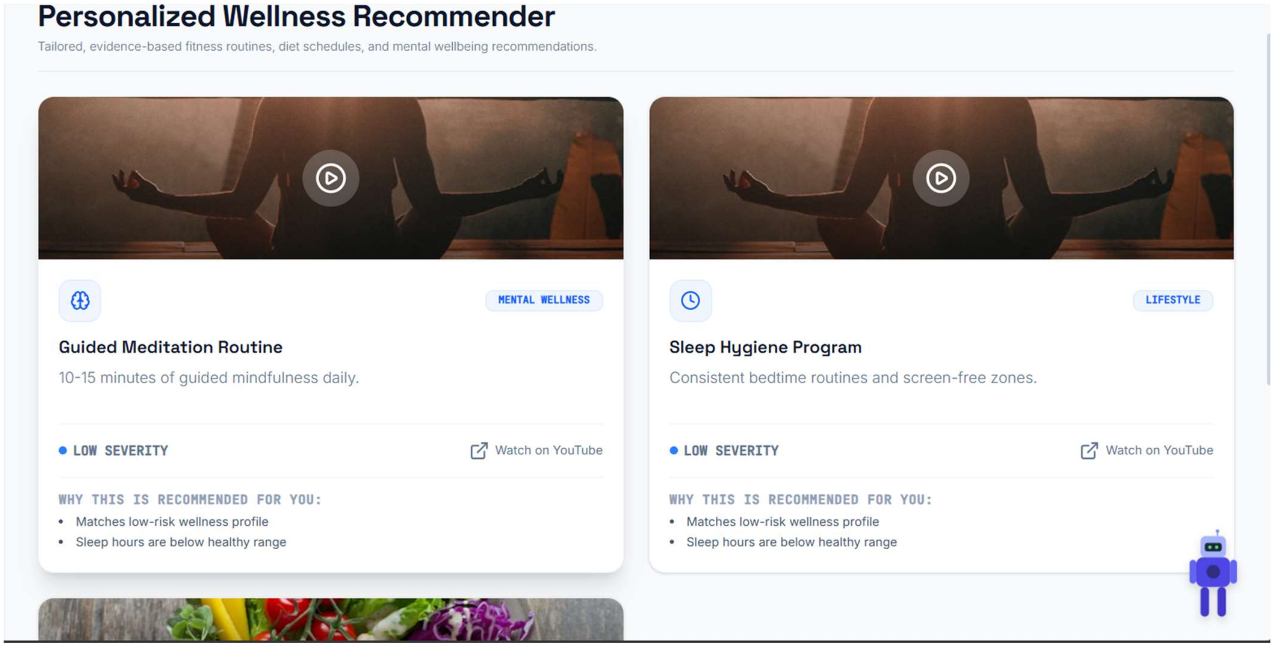
Risk Prediction



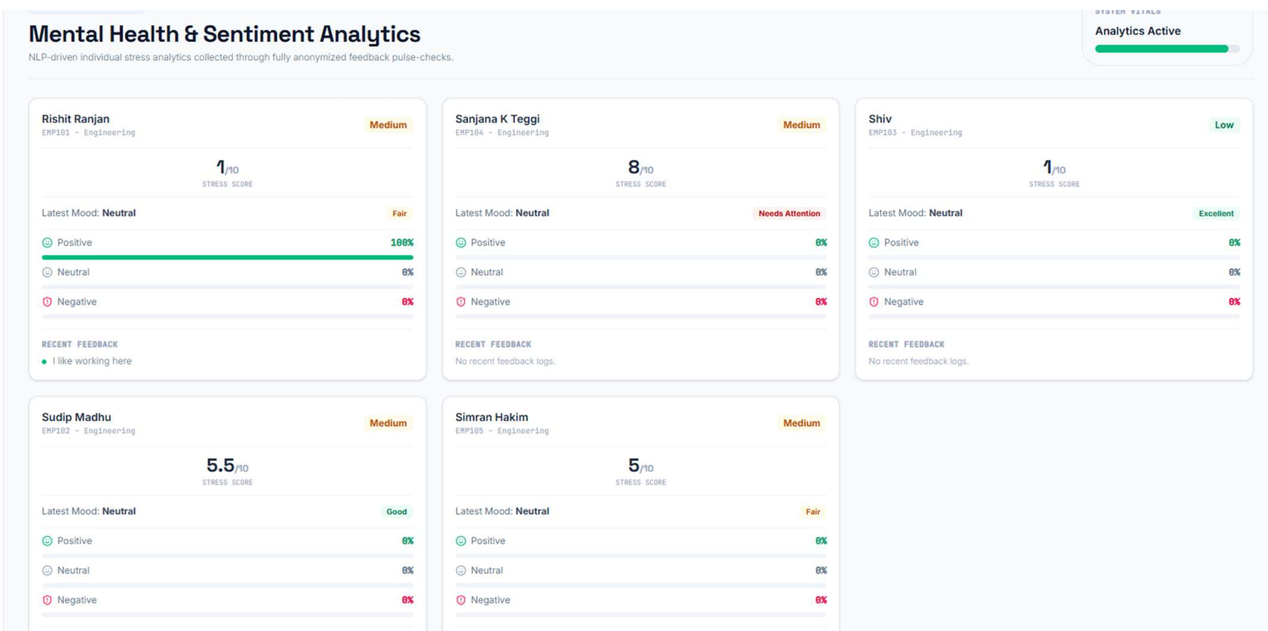
Wellness Recommendation System



Personalized Wellness Recommendation for Employee

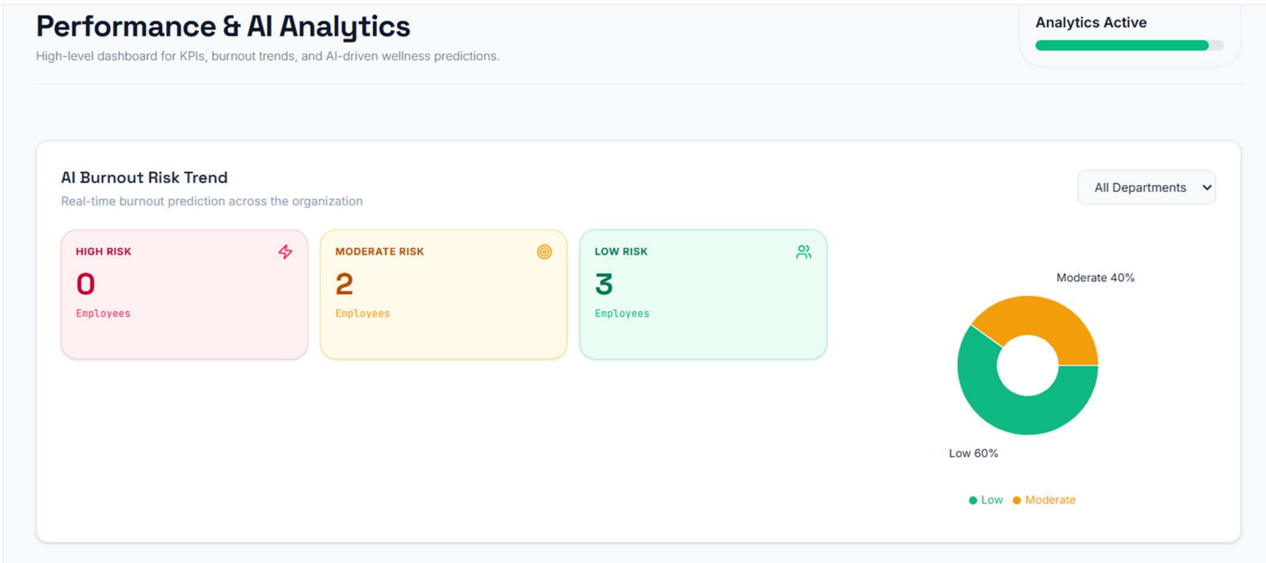


Sentiment Analysis of each Employee

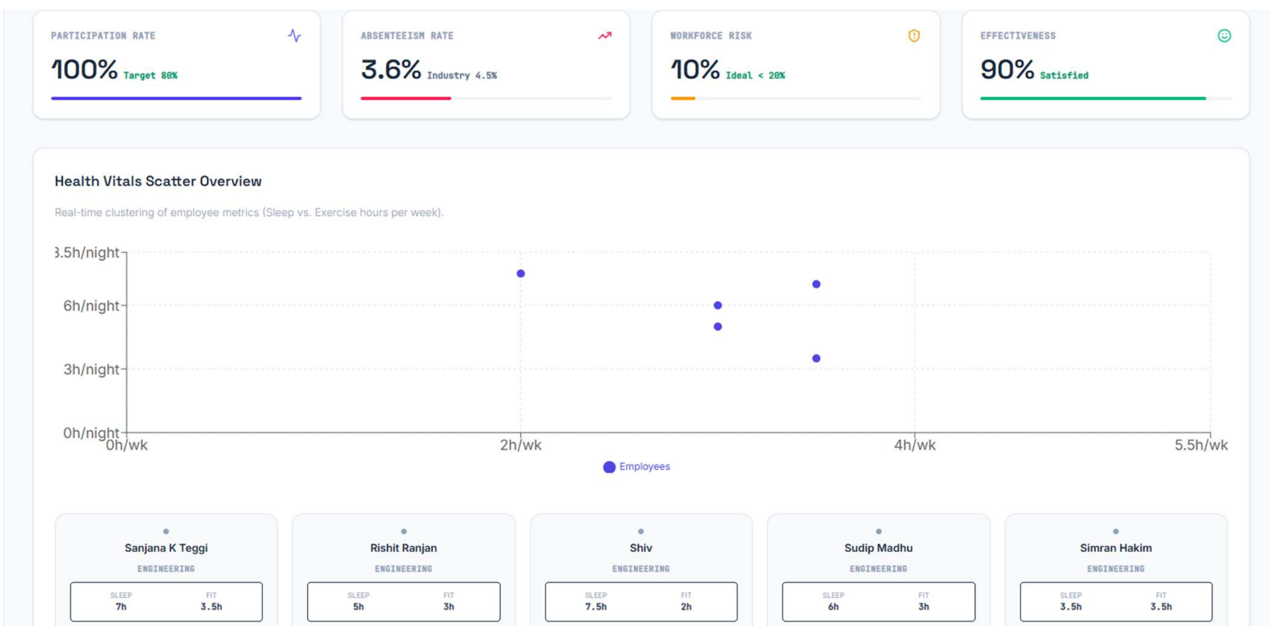


Performance & AI analytics

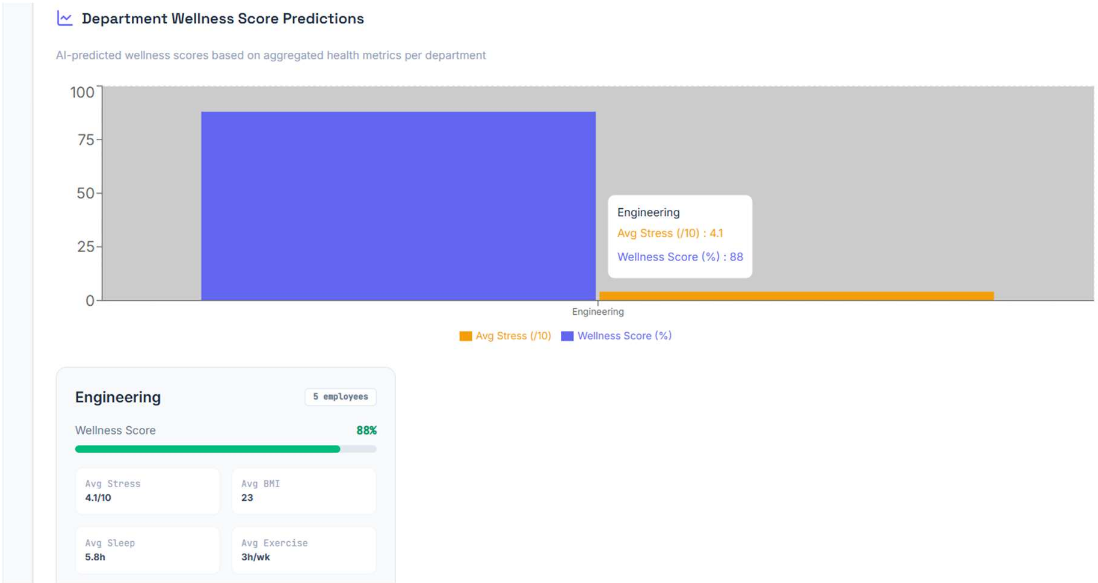
AI Burnout Risk



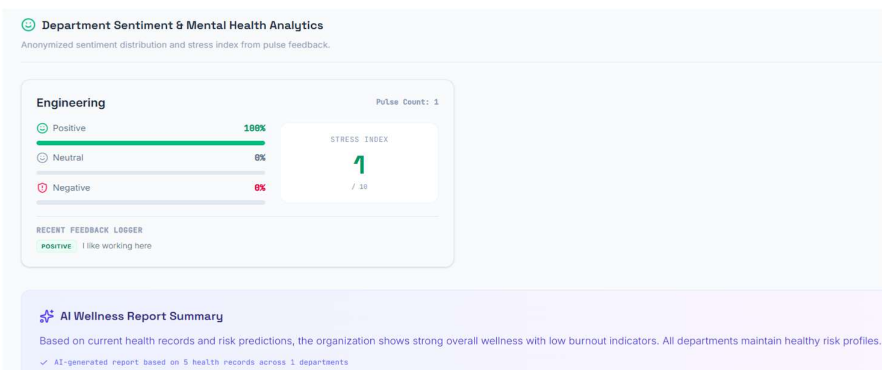
Health Vitals Scatter Plot



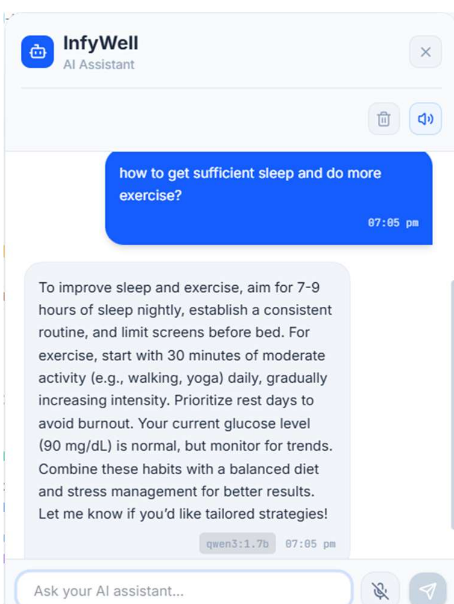
Department Wellness Score Prediction



Departmental Sentiment & Mental Health Analytics



AI Floating Chatbot with voice assistance

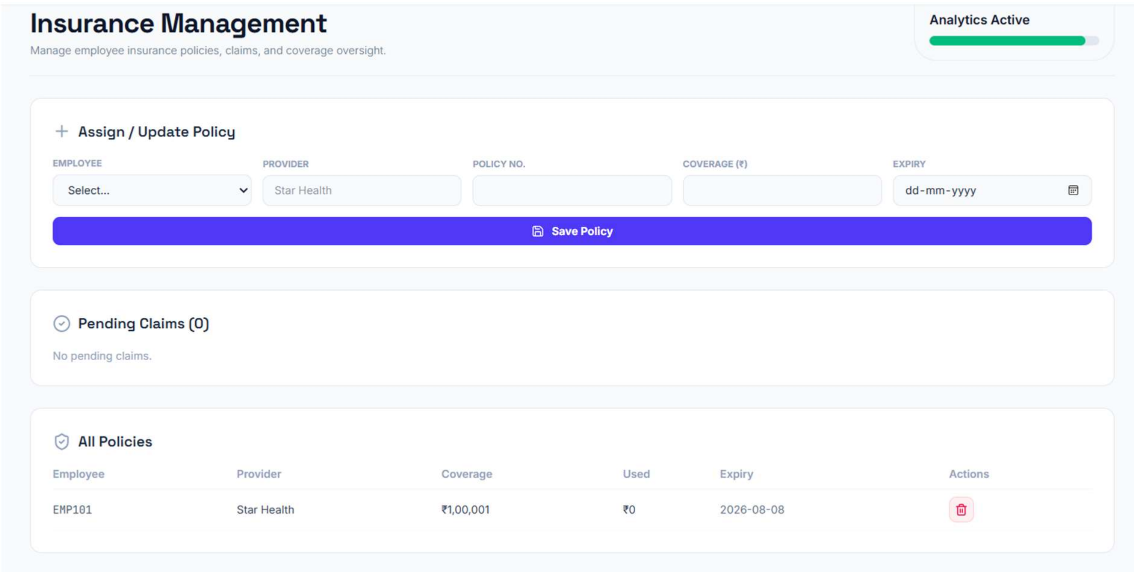


Milestone 4: Weeks 7–8 — Extensions and Finalization

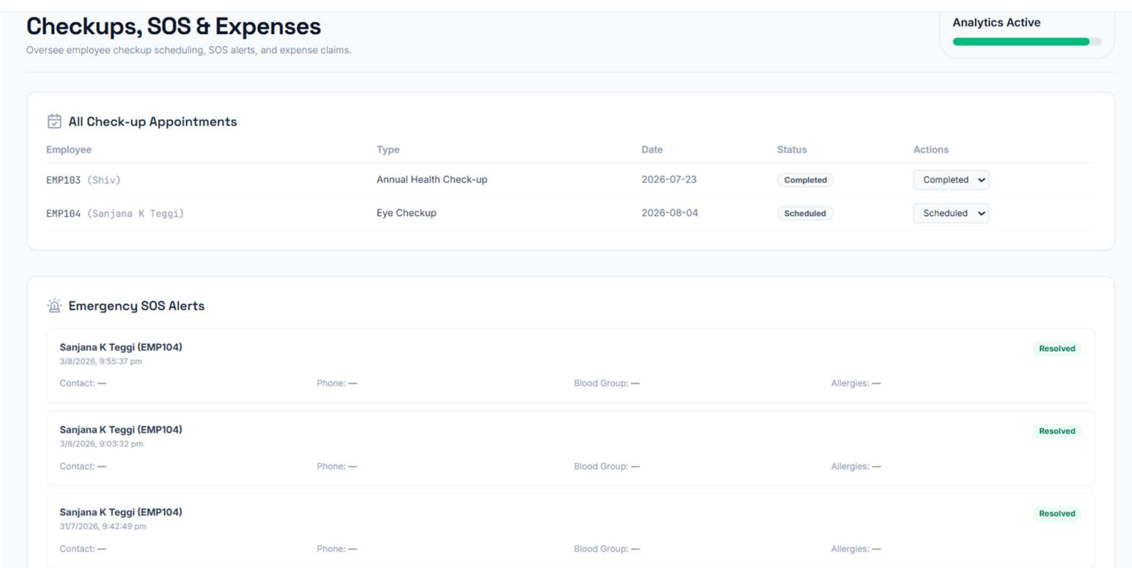
Goals: Add notifications, insurance, check-ups, SOS, expense modules, testing, and documentation

Tools: React components, Flask endpoints, MongoDB, deployment tools

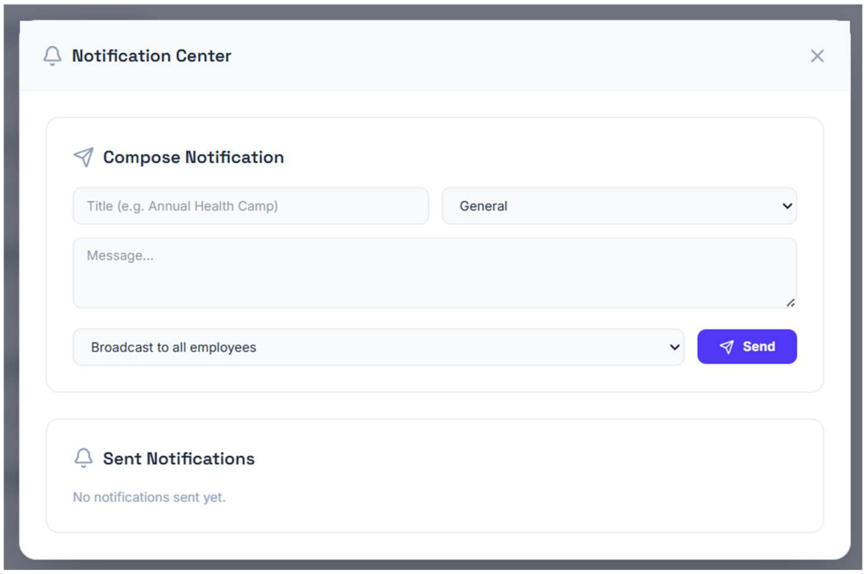
Insurance Management



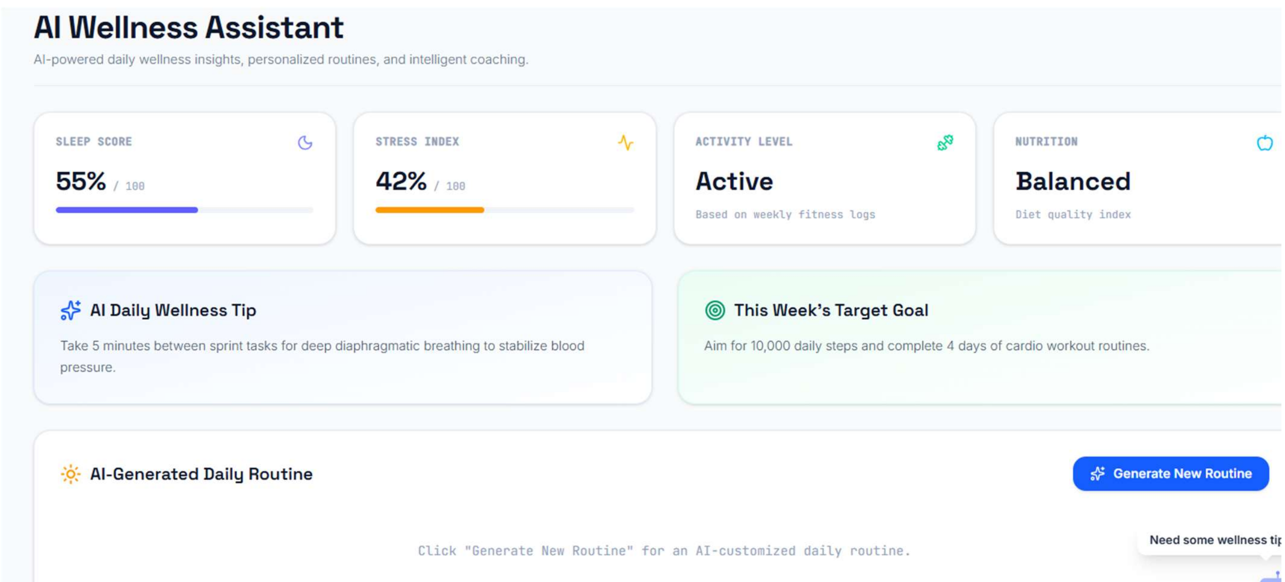
Checkups, SOS & Expenses



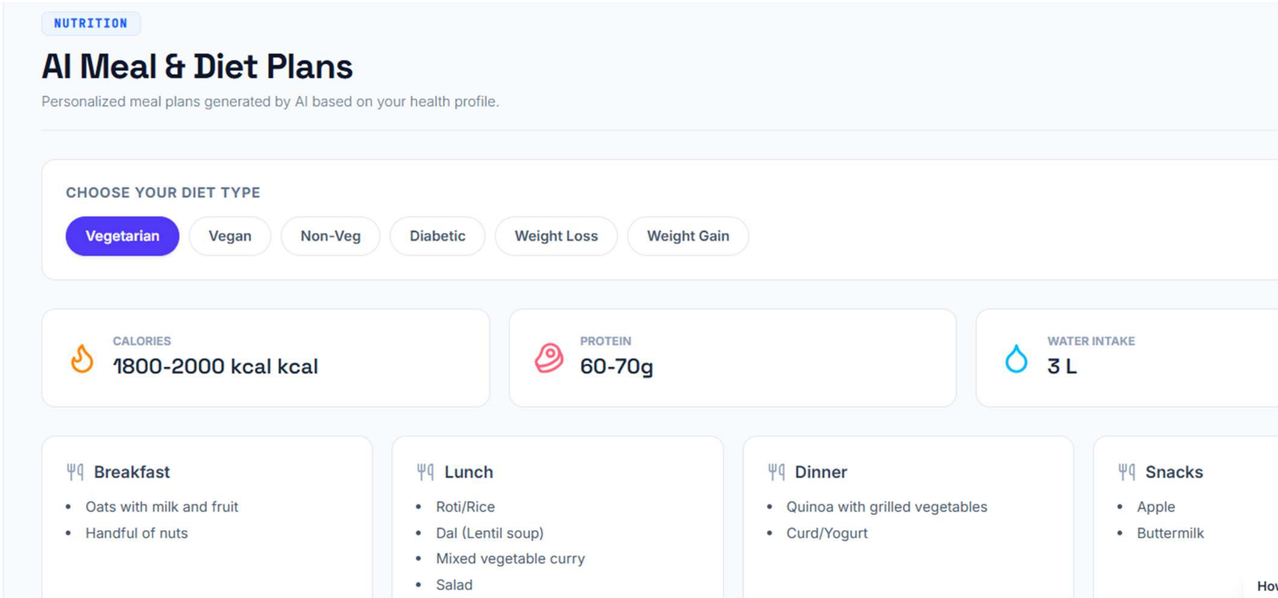
Notification Center



AI Wellness Assistant for Routine Generation



AI Meals and Diet plan based on health profile



Deliverables

- Working employee and admin dashboards.
- Persistent MongoDB-backed wellness records.
- Risk, recommendation, sentiment, and KPI modules.
- Extension modules and final documentation.

5. Evaluation Criteria

Milestone	Success Criteria
Week 2	Registration, login, password recovery, role-based access, and profile flow operate correctly.
Week 4	Health records and employee wellness profiles can be created, updated, stored, and displayed.
Week 6	Risk prediction, recommendations, sentiment analysis, chatbot, and KPI modules produce usable outputs.
Week 8	Extension modules, testing, documentation, and deployment preparation are complete.

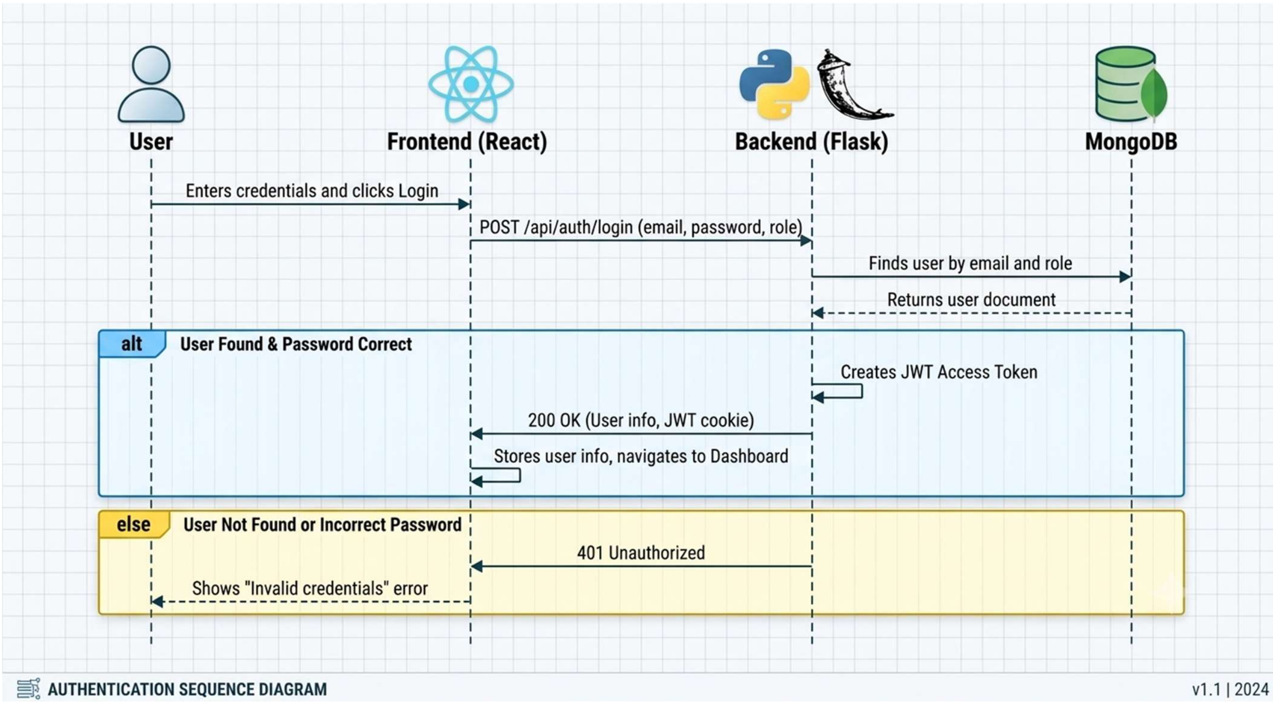
Testing Focus

- Authentication success and failure cases.
- Authorization differences between employee and admin roles.
- Validation of health values and duplicate health-record prevention.
- Persistence and synchronization of dashboard data.
- Correct risk classification and readable recommendations.
- Responsive interface and clear error or success feedback.

6. Flow Diagram

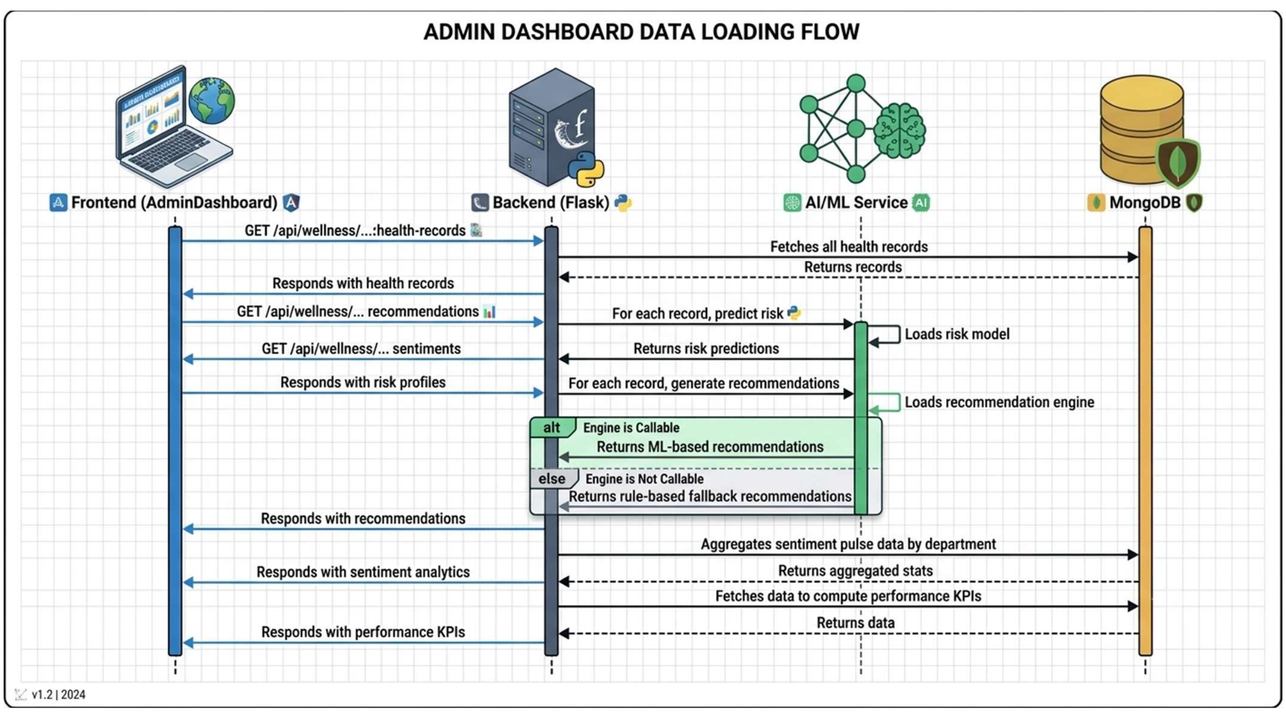
User Authentication Flow

This flow describes how a user logs in and gains access to the dashboard.



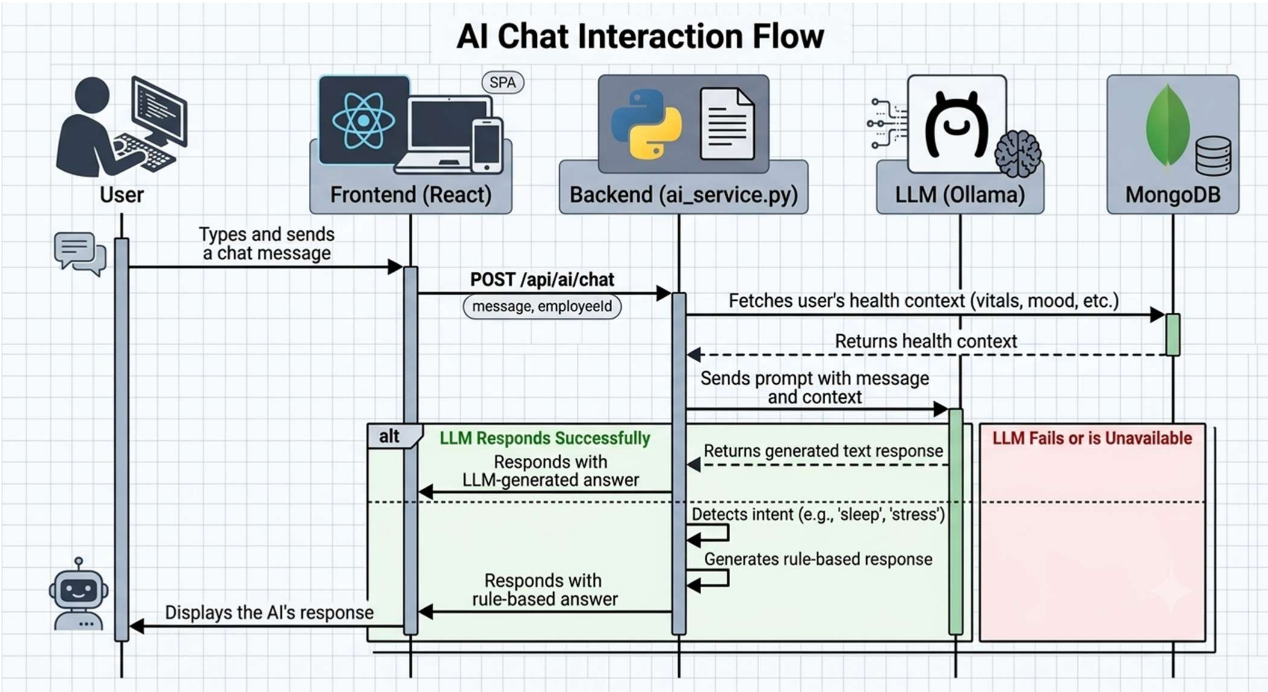
Admin Dashboard Flow

This flow outlines how the Admin Dashboard populates its various analytics sections upon loading.



AI Chat Interaction Flow

This flow shows how the AI chat feature processes a user's message, attempting to use a Local Large Language Model (LLM) and falling back to a rule-based system if needed.

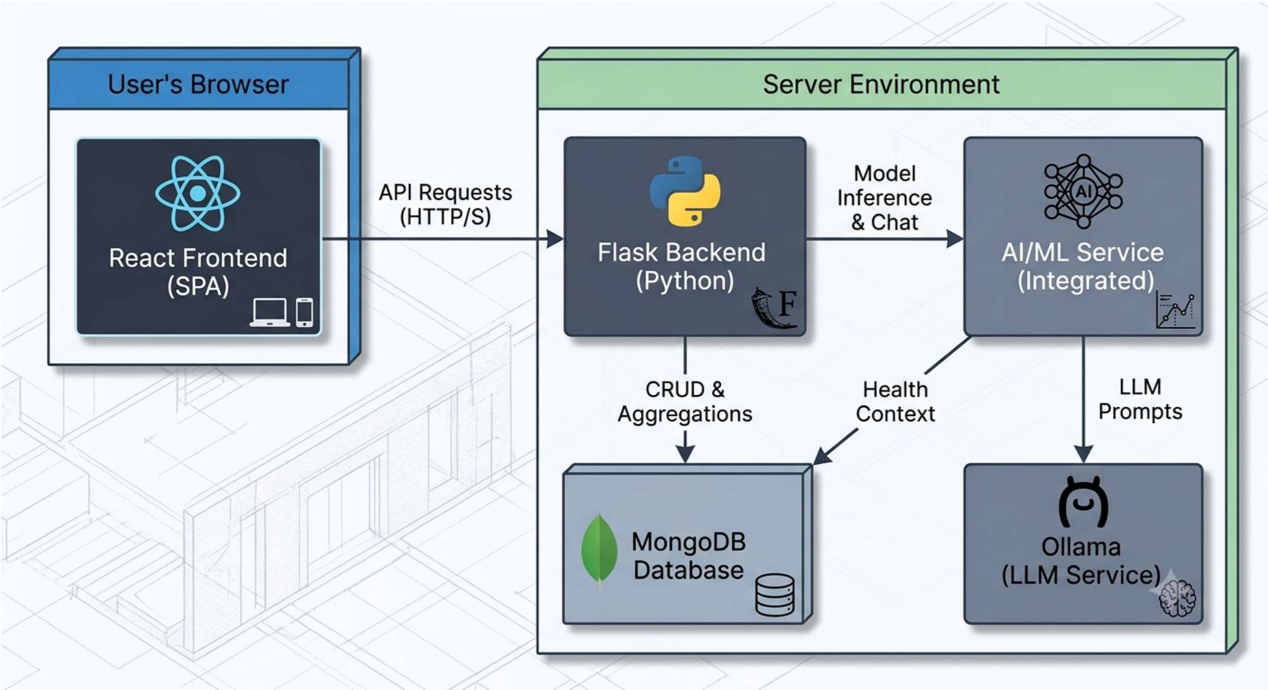


Login / Signup	Role Check	Dashboard	Wellness Data	AI & Analytics	Action
Secure access	Employee/Admin	Role modules	Vitals, habits, mood	Risk, recommendations, KPIs	Feedback and updates

Employee Flow	Admin Flow
Login ->Profile -> Health Vitals -> Risk Diagnostic -> Recommendations -> Feedback / Chatbot	Login -> Admin Dashboard -> Health Data -> Risk Prediction -> Sentiment / KPIs -> Notifications and Extensions

7. System Architecture

The application is built on a modern, decoupled architecture, commonly known as a Backend for Frontend (BFF) pattern. This means the React frontend is a distinct application that communicates with a dedicated Flask backend via a REST API.



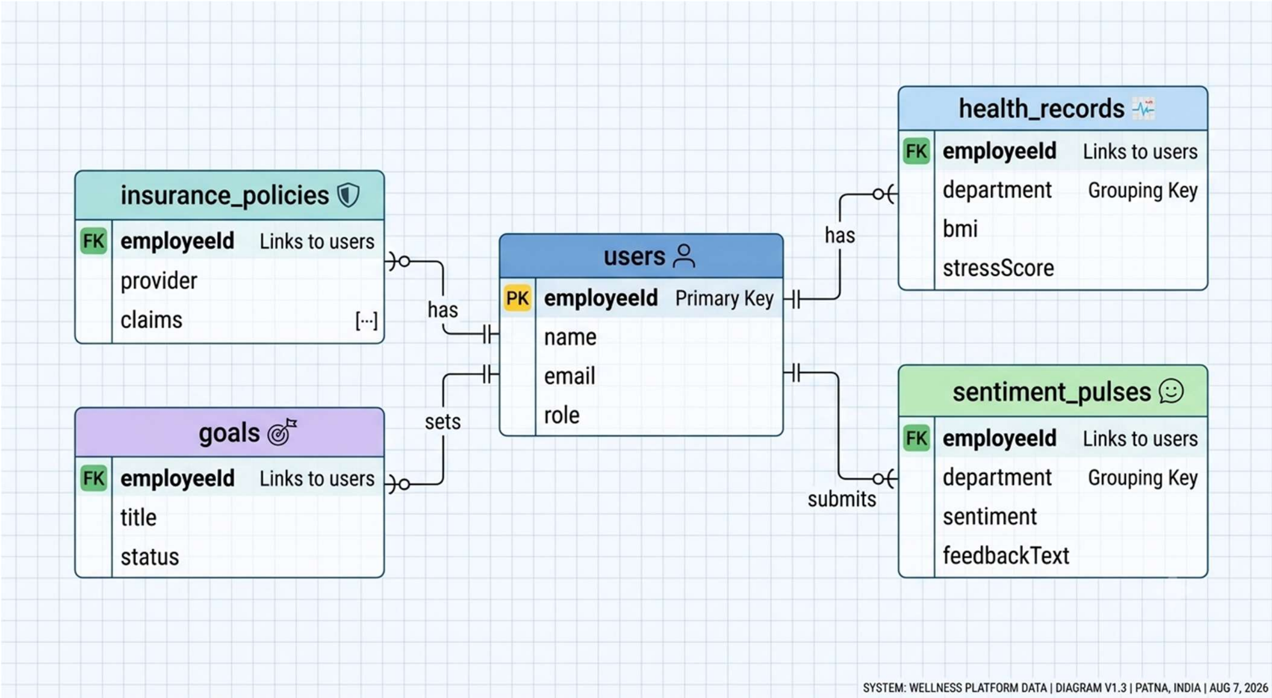
Frontend Layer	Backend Layer	Data and Intelligence Layer
React + Vite	Flask API	MongoDB
Login, Signup, Dashboards, Forms	Authentication, validation, REST endpoints	Users, admin, records, habits, logs, sentiments
Employee/Admin interaction	JWT cookies, bcrypt, CORS, file upload	Risk model, recommendation engine, NLTK sentiment analysis

Request and Response Flow
User action -> React component -> Flask REST endpoint -> MongoDB/model processing -> JSON response -> dashboard state update.

8. Schema Diagram and Data Design

Collection	Important Fields	Purpose
admin	_id, adminId, name, email, passwordHash, role, createdAt	Administrator accounts and access permissions
checkup_appointments	_id, employeeId, doctorName, appointmentDate, status, notes	Scheduled routine health checkups and medical visits
daily_habits	_id, employeeId, waterCups, stepsCount, mood, streakDays, date	Daily activity tracking and habit streaks
employee_feedback	_id, employeeId, category, message, rating, createdAt	General wellness feedback and platform reviews
goals	_id, employeeId, title, targetValue, currentValue, unit, deadline, status	Personal fitness and health targets
health_expenses	_id, employeeId, claimAmount, category, receiptUrl, status, submittedAt	Medical claim requests and health expense reimbursements
health_history	_id, employeeId, condition, diagnosedDate, medications, allergies	Longitudinal health trends and historical medical records
health_records	_id, employeeId, department, bmi, bloodPressure, exerciseHoursPerWeek, sleepHoursPerNight, stressLevel, healthAssessment	Primary physical vitals and health diagnostic metrics
insurance_policies	_id, employeeId, policyNumber, provider, coverageAmount, expiryDate	Employee health insurance coverage details
mental_health_logs	_id, employeeId, stress, feedback, moodScore, createdAt	Mental wellness entries and stress tracking
notifications	_id, employeeId, title, message, isRead, type, createdAt	System alerts, reminders, and broadcast messages
password_reset_requests	_id, email, otp, resetToken, expiresAt, used	Password recovery and token management
sentiment_pulses	_id, department, averageStressScore, sentimentDistribution, keyIssues, timestamp	Aggregated department sentiment and workplace pulse analytics
sos_alerts	_id, employeeId, location, emergencyContact, status, triggeredAt	Critical health/emergency notifications and immediate alerts
system_settings	_id, settingKey, settingValue, updatedBy, updatedAt	Global application parameters and module flags
users	_id, employeeId, name, username, email, passwordHash, role, avatarUrl, createdAt	Core user directory, authentication, and profiles

Schema Diagram



Conclusion

In conclusion, the Employee Wellness Management Analytics Platform successfully integrates a secure, full-stack architecture with intelligent, data-driven features. By leveraging a React frontend, a robust Flask backend, and a sophisticated AI/ML service, the application delivers a seamless and insightful user experience. It effectively addresses the critical need for proactive wellness management by providing distinct, role-based dashboards for both employees and administrators.

Key outcomes, such as predictive risk analysis, personalized AI recommendations, and comprehensive sentiment analytics, have been fully realized. The platform moves beyond generic solutions, offering a personalized and engaging ecosystem for health monitoring. The extensible REST API and modular design ensure the system is not only functional but also scalable for future enhancements. Ultimately, this project stands as a powerful proof-of-concept for how technology can be used to foster a healthier, more resilient, and productive workforce.